Scale, Asymmetry, and the Structure of Distinction
An Essay

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Abstract

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The arithmetic we use to count apples and sheep has become a time bomb now that we apply it to the quantum world and the cosmos. This essay argues that the foundational crisis in contemporary physics and mathematics stems not from insufficient data or computational power, but from the uncritical application of a scale-local language—classical arithmetic—to ontologies operating at radically different regimes of distinction. We propose a methodological shift: treating *scale* not as an external metric, but as an ontological operator that structures the very possibility of objects, numbers, space, and the observing "I."

Developing a framework of *operational phenomenology*, we identify a fundamental asymmetry along the scale axis: a rigid lower limit (where distinction dissolves into fluctuation) and an unbounded upper horizon (where organization persists). This asymmetry is shown to be a precondition for any act of distinction, not a derivative physical property. Within this frame, we reinterpret key cosmological phenomena—dark energy, the cosmological horizon, the Big Bang—not as physical substances or events, but as effects of transitions between regimes of distinction. We further sketch a formal calculus based on a non-commutative scale operator and a graph-theoretic algebra of forms, positioning arithmetic as a secondary, scale-stabilized dialect of a more primary geometric-topological language.

The essay contributes a meta-language for foundational inquiry, bridging phenomenology, the philosophy of mathematics, and theoretical physics. It offers falsifiable criteria, connects to existing programs (renormalization group, relational quantum mechanics, topos theory), and opens pathways for rethinking quantum gravity, consciousness, and artificial intelligence as questions about the boundaries and stability of distinction itself.

Keywords: ontology of scale, operational phenomenology, foundations of mathematics, asymmetry, distinction, quantum gravity, cosmology, category theory

"The arithmetic we use to count apples and sheep has become a time bomb now that we apply it to the quantum world and the cosmos."

Here is a simple question that might seem naive even to a schoolchild: where exactly does "arithmetic" cease to apply?

At first glance, the answer is obvious. Arithmetic works everywhere—from counting apples to calculating cosmic trajectories. What could be more reliable than $1 + 1 = 2$? Mathematical operations are symmetric, reversible, deterministic, and universal. Yet it is precisely this obviousness that proves to be a trap.

Upon deeper investigation, a strangeness emerges. We know that there is uncertainty, randomness, probability—things that clearly do not fit within the framework of classical arithmetic. Somewhere between counting apples and quantum mechanics, a rupture occurs. But where exactly?

The next step leads to an even deeper perplexity. The number π, the circle, and the square belong to different ontologies. The square lives in the world of linear equations; the circle, in the world of transcendental functions. The attempt to "square the circle" is not a technical problem but a collision of two languages for describing reality.

Stop!—one might say.

> *The arithmetic we use for apples and sheep became a time bomb when we began applying it to the quantum world and the cosmos.*

This is not a metaphor. This is a diagnosis.

What is happening? What is happening is that we continue to employ the same basic language—the same operations, the same numbers—to describe radically different ontologies. Yet we do not rethink arithmetic itself. We simply layer new tools atop the old, creating an illusion of progress.

This holds even when dealing with large numbers, infinities, or infinitesimals. It is then that we recall the concept of scale.

Scale is not merely "size" or "unit of measurement." It is a parameter that determines the very structure of distinction—that which enables the appearance of objects, numbers, space, and even the "I" itself.

Methodological Framework

Our investigation begins by rejecting the conventional notion of scale as an external measure. Instead, scale is treated as an operator that determines the structure of the very act of distinction—that which enables the appearance of objects, numbers, space, and even the "I" itself.

Key stance: rejecting binary oppositions and the "nesting-doll" hierarchy of worlds in favor of their juxtaposition. The investigation proceeds within the tradition of operational phenomenology—an approach that employs the phenomenological method to access structures subsequently discovered to be universal.

Below, we outline the ideas we find most original and promising.

Fractals and Anti-Fractals

For a long time, fractals were regarded as a universal model of reality's structure. Self-similarity at all scales promised symmetry and harmony: look closer, and you see the same thing; move farther away, and again the same thing.

But reality proves otherwise.

In the microworld, fractal logic breaks down. We encounter a threshold beyond which "looking closer" loses its meaning—the uncertainty principle, quantization, possibly the discreteness of spacetime. The world at small scales is not fractal in nature. Discrete networks, probabilistic clouds, nonlocal correlations—none preserve self-similarity under scaling.

This is the first indication of a fundamental asymmetry along the scale axis: in one direction, the potential for infinite repetition; in the other, a rigid limit.

The Tripartite Structure of Distinction: Field, Point, Boundary

In immediate experience, we discover a tripartite structure: the field, the point of reference ("here"), and the boundary that makes this distinction possible while remaining invisible from within.

When we "change scale," we alter the entire triad:

> - The field ceases to be a homogeneous background—new inhomogeneities appear within it, or, conversely, previous structures "dissolve." - The point of reference shifts—what was an "object" becomes "background," and vice versa. - The boundary is redrawn—what was "inside" perception recedes beyond the horizon, while new elements intrude from "outside."

To avoid anthropocentrism, we adopt the following approach: the tripartite structure is first registered in self-observation. It is then found to be reproduced in other systems. Colliding billiard balls do not "see" each other's interiors; their interaction entails a "blind spot" (mass distributed throughout the volume). A quantum particle is subject to a fundamental limit on the simultaneous measurement of position and momentum—this is not an instrumental limitation but a property of reality itself.

We are dealing with an ontological invariant, not an anthropocentric projection.

The "I" as Singularity of Distinction

The phenomenon of the "I" (pure presence, the point of reference) serves as the basic model of the field of distinction. Its structure—the boundary between "I" and "not-I," the absence of spatial localization of this boundary, the asymmetry of scale—is reproduced at all levels of reality.

In experience, the first thing that emerges from the pure experience of the "I" is a limit—not a line on a map, but a sensation: "further—cannot." This limit has no fixed location; it is everywhere and nowhere, like the center of a sphere that coincides with every point on its surface.

The image of the "I" as a wormhole (cf. Leibniz's monads) arises from the following observation: the "I" is a point through which infinite worlds pass, yet the point itself belongs to none of them. The monad—a mathematical point behind which lies infinity—here acquires operational meaning: the "I" is such a point, and upon a change of scale, it can unfold into entire worlds or collapse into pure being.

The Asymmetry of the "I"

The asymmetry of the "I" is analogous to that of the scale axis. Below (the extremely small), there is a rigid limit: an atom does not possess an "I"; an

electron does not reflect. There is a minimum scale below which the "I" dissolves into non-distinction.

Above (expansion), there is no such limit: one can imagine the "I" of a planet, a galaxy, the universe. The limit here is practical rather than ontological.

This asymmetry is not derived from other principles; it is given as a condition for the possibility of distinction itself. Without it, there would be neither objects nor an observer.

The Dual Meaning of "I" and a Methodological Clarification

To avoid accusations of anthropocentrism, it is necessary to distinguish two senses of "I."

First sense: "I" as phenomenological givenness—that which is directly given in experience: the point from which perception is conducted. This sense is tied to a specific observer.

Second sense: "I" as universal principle of distinction—the structural position that arises wherever an act of distinction occurs. This could be termed the "point of reference"—that relative to which the field is structured, divided into "here" and "there."

The transition from the first sense to the second is not the result of phenomenological reduction in the strict Husserlian sense, but a conscious adoption of a metaphysical hypothesis—*operational realism*. We use human experience as empirical access to a structure subsequently discovered to be universal. This is not panpsychism (we do not ascribe "consciousness" to a stone or an electron), but rather the claim that the structure of distinction, which in humans is experienced as the "I," can be described in other systems in terms of relations that do not presuppose subjectivity.

Form and Geometry Are Prior to Arithmetic

What is form?

Form is a wholeness given prior to any distinction into parts. It is not a geometric figure, but rather something that has not yet dissolved into objects and background. Form changes with a change of scale, but this is a change of the whole, not merely the appearance of parts.

When we "change scale," we do not single out objects. We alter the form of the field itself:

> - At one scale, the field appears smooth and continuous. - At another, folds, curves, and irregularities appear within it. - At a third, these irregularities acquire their own structure, their own geometry.

Asymmetry of form

When decreasing scale, form does not simply become more detailed—it disintegrates: a smooth surface turns into noise, into chaos, into something that can no longer be called "form."

When increasing scale, form does not disintegrate—it organizes. New levels of structure emerge, yet wholeness is preserved.

This is an asymmetry not of counting, but of form.

Boundary

The boundary is not a line between objects. It is the limit of form. The point beyond which form ceases to be form. And this boundary is asymmetrical: in one direction, disintegration; in the other, organization.

When Does Countability Emerge?

Countability ($1 + 1 = 2$) presupposes stable, isolated objects. However, objects arise only when the boundary acquires a certain structure—when the limit becomes "smooth" and reproducible within a certain range of focal lengths.

Outside this range, there are no objects, no counting. Not because "the laws are different," but because the conditions for the very phenomenon of the object to arise are absent.

Arithmetic is not a universal language of reality. It is a local dialect, operative only in that mode of attention where the boundary stabilizes in a certain configuration.

Geometry (form, topology, structure) is primary: it describes how the field of distinction is organized prior to the emergence of countable objects within it. A change of scale is a transformation of form, not a change in the number of objects.

Caution Regarding "New Mathematics"

The phrase "new mathematics" might create the false impression that we are proposing a ready-made alternative formal system. This is not the case. Our goal is not to construct a new axiomatic, but rather a *meta-language for the foundation of mathematics*—a phenomenology of mathematical experience. We investigate:

> - Why is arithmetic possible at all? - Why does it work only within a certain range of scales? - Why does it "break" outside this range (the quantum world, cosmology)?

The proposed axioms and the sketch of a graph algebra are not an alternative mathematics, but rather a *conceptual framework* within which existing mathematics can be understood as a particular, scale-limited regime.

Operational Grounding

The experience of consistently changing the scale of attention reveals:

1. The field of distinction exists prior to the isolation of objects. At one scale, the field appears homogeneous; at another, structures emerge within it. This accords with the definition of form as a wholeness preceding dissolution into objects.

2. The order of scale change matters—zooming in and zooming out are asymmetrical. Zooming in, we sooner or later encounter a limit; zooming out, no such limit exists.

3. In one direction ("inward"), there is a rigid limit of distinction—the structure disintegrates into fluctuations, instability. In the other direction ("outward"), there is no limit—the object simply leaves the field of view, yet the field remains distinguishable.

Asymmetry as a Primary Ontological Principle

Asymmetry (a rigid limit when moving "inward," no limit when moving "outward") is not a property of the world explicable by physics. It is a condition for the possibility of any distinction whatsoever. Without it, there would be neither objects nor an observer.

Our position is *operational realism*: the asymmetry of scale is a property of reality that manifests in all systems and is accessible to any act of distinction. The problem of quantum gravity thus appears not as the search for a unified theory, but as the study of transitions between regimes of distinction.

Cosmological Implications

Cosmology is not the science of "the universe as an object," but the study of regimes of distinction at cosmological scales.

Dark energy is not a "substance," but an effect that arises when transitioning to a scale where the structure of distinction changes its mode. At larger scales, gravity weakens, and a new principle (repulsion) emerges, one that has no analogue in the microworld.

The cosmological horizon is not "what we do not see," but the boundary of the regime of distinction: beyond it, different principles operate. The question of what is "there" is ill-posed, because there are no conditions for distinguishing "there" from "here."

The Big Bang is not "the beginning of time," but the limit of limitation itself: the point beyond which distinction is impossible. The question of what was "before" is meaningless, because there are no conditions for distinguishing "before" from "after."

The arrow of time is not a consequence of entropy, but a manifestation of the asymmetry of scale: time "flows" in the direction of increasing scale, where new levels of organization open up; in the direction of decreasing scale, time "collapses" (quantum processes are reversible).

The fine-tuning of constants is neither chance nor design, but the conditions for the possibility of the field of distinction in the mode we call "our universe." Other constants would entail another mode, in which distinction is either impossible or operates differently.

Phenomenological Interpretation vs. Physical Model

The proposed understanding of cosmological phenomena does not claim to supplant physical theories. It does not derive the equation of state for dark energy, nor does it predict the value of the cosmological constant. Its task is different: to indicate what kind of ontological shift could render the observed effects natural rather than paradoxical.

Physicists may continue to work with the ΛCDM model, loop quantum gravity, etc. Our interpretation offers them a meta-language: it explains why effects arise at cosmological scales that have no analogue in the microworld.

Operationalizing Asymmetry: From Metric to Topology

Introducing the parameters λ_{min} and λ_{max} allows us to fix asymmetry as a relation of limits, yet they presuppose a numerical metric. For a "numberless" phenomenological mathematics, a method of assessing asymmetry through topological invariants is proposed.

Clarification: ordinal vs. cardinal numbers

It is important to distinguish two senses of "number":

- **Cardinal numbers**—the result of counting stable objects, a secondary phenomenon.
- **Ordinal numbers**—the result of iteration ("first step," "second step"), ontologically prior to counting.

The parameters n_{min} and n_{max} refer precisely to ordinal numbers. We do not "count" objects; we fix how many times the approximation operator can be applied before reaching a limit.

Asymmetry as number of levels of connectivity

Instead of the metric λ, one can operate with the order of connectivity—the number of levels at which the structure preserves topological properties. By successively applying the operator $\Gamma(\lambda)$, we record:

- n_{min}—the number of steps after which the topological invariant ceases to be defined. - n_{max}—the number of steps when increasing scale (which may be infinite).

Asymmetry is expressed as $n_{max} \to \infty$ with finite n_{min}.

Graph approach

The field of distinction can be represented as a graph, where nodes are stable inhomogeneities and edges are the relations between them. When decreasing scale, nodes break down into subnetworks, losing connectivity; when increasing scale, they merge into clusters.

Empirical examples:

- Human perception: the visual field, when zoomed in, ceases to be connected at the pixel level ($n_{min} \sim 5$–10); when zoomed out, connectivity is preserved at the levels of room, house, city ($n_{max} \to \infty$).
- Quantum system: the wave function, when approaching the Planck scale, loses its metric structure (n_{min} finite); when zoomed out, it becomes more complex, transitioning to the classical world ($n_{max} \to \infty$).

Sketch of a Formal Calculus for the Scale Operator

Basic objects

- **Field** Φ—unstructured givenness, prior to any distinction.
- **Scale operator** $\Gamma(\lambda)$—an action that, at a given mode λ, distinguishes forms from the field.
- **Form** $F(\lambda) = \Gamma(\lambda)[\Phi]$—the result of applying the operator: a holistic configuration distinguishable at the given λ.

Axioms of the calculus

1. **Non-commutativity:** $\Gamma(\lambda_2) \circ \Gamma(\lambda_1) \neq \Gamma(\lambda_1) \circ \Gamma(\lambda_2)$. The order of scale change matters; zooming in and zooming out are irreversible.

2. **Monotonicity:** If $\lambda_1 < \lambda_2$ (λ_1 is "deeper"), then $F(\lambda_1)$ contains more detail than $F(\lambda_2)$.

3. **Existence of a lower limit:** There exists a minimal mode λ_{min} below which $\Gamma(\lambda)$ is not defined.

4. **Absence of an upper limit:** For any λ, there exists $\lambda' > \lambda$, and $F(\lambda')$ contains a less detailed but still distinguishable structure.

Operations on forms

- **Comparison:** $F_1 \prec F_2$ means that F_2 is obtained from F_1 by enlargement.

- **Composition:** If $F_2 = \Gamma(\lambda_2)[F_1]$ and $F_3 = \Gamma(\lambda_3)[F_2]$, then $F_3 = \Gamma(\lambda_2 \cdot \lambda_3)[F_1]$.

- **Invariants:** A topological property $I(F)$ (connectivity, number of components) is an invariant of distinction if $I(F(\lambda_1)) = I(F(\lambda_2))$ for all λ in some range.

Simplest implementation: a graph algebra of distinction

Let Φ be an unstructured set of "points." The operator $\Gamma(k)$ (k — grouping level) acts as follows:

- $F(0)$—each point is isolated.

- $\Gamma(1)$ groups points into clusters.

- $\Gamma(2)$ groups clusters into larger blocks.

Then:

- When decreasing k, we return to a finer structure.

- When increasing k, we obtain more abstract forms.

Invariant: the number of connected components c_k.

Asymmetry: c_k as $k \to \infty$ tends to 1; as $k \to 0$, the structure hits discrete points—a limit beyond which there are no new levels.

Here k is not a numerical metric, but an ordinal number of steps.

Application to "Hard Problems"

Quantum gravity: the Planck scale is the boundary of distinction, beyond which spacetime loses metric meaning.

Consciousness: asymmetry explains why consciousness is impossible below a certain size (a neuron) and why it is possible above (collective consciousness). Below—a rigid threshold: an atom does not possess an "I." Above—no fundamental prohibition, but practical limitations (speed of light, thermodynamics).

Artificial intelligence: the question is not "does AI have consciousness?" but "at what scale of complexity does a regime analogous to the point of reference emerge?"

(Cf. Giulio Tononi's Integrated Information Theory (IIT), where consciousness is linked to the quantity Φ—a measure of a system's integrated information. Unlike IIT, where consciousness is defined by a quantitative threshold, our framework emphasizes a topological transition: the emergence of the point of reference occurs when the form of distinction stabilizes at a certain scale of complexity.)

Connection with Existing Research

The proposed framework is not isolated. It resonates with:

The renormalization group in quantum field theory, where effective laws change with scale. Our ontology offers an ontological grounding for this phenomenon: the asymmetry of UV and IR limits is not an empirical fact but a manifestation of a fundamental property of distinction.

Relational quantum mechanics (Rovelli), where a state exists only relative to another system. Our framework concurs with the primacy of interaction but goes further, deriving from scale asymmetry not only quantum effects but also the foundations of mathematics.

Lawvere's program (ETCS) and topos theory, where objects are defined through relations rather than internal structure. The field of distinction can be understood as a topos, and different scales as different topoi with distinct internal logics.

Constructivism in its critique of Platonism, but unlike it, we do not reduce mathematics to subjective constructions; rather, we situate it in the field of distinction, which has an objective character.

Falsifiability Criterion

The hypothesis of ontological scale asymmetry makes predictions about what *cannot* be discovered:

- One cannot discover a system whose structure, under arbitrarily deep approximation, is preserved without a change of ontology.
- One cannot discover a system whose structure, under arbitrarily far distancing, disappears (reaches a finite limit).

If such a system were discovered, the hypothesis would be refuted.

Conclusion: Findings and Prospects

The following conceptual nodes emerge from the analysis:

1. **Asymmetry of scale as a fundamental ontological principle**, a condition for the possibility of distinction itself, not a consequence of physical laws.
2. **The primacy of form and geometry over arithmetic.** Arithmetic is a particular, scale-limited regime of the field of distinction.

3. **The "I" as an operator of distinction and prototype of scale structure**, linking the phenomenology of consciousness to the ontology of scale.

4. **Cosmology as the study of regimes of distinction at extreme scales**, redefining the status of dark energy, singularity, and horizon.

5. **Operational phenomenology as a method**, allowing the foundations of mathematics and physics to be built from the immediate phenomena of distinction.

Directions for further work:

- *Operationalization:* transition from metric to topology (n_{min}, n_{max} as ordinal indices, graph approach).

- *Application to "hard problems":* quantum gravity, consciousness, AI—as questions about the boundaries and regimes of distinction.

- *Formalization:* developing the axiomatics in the direction of category theory and topoi.

The analysis presented draws on the Ontology-Lab project (https://github.com/SergeakaAimate/Ontology-Lab).

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